

Unit

2

Lesson

1

Innovators in Action.

The Brain Behind the System.

Time Frame: [90 min]



Lesson Objectives:

By the end of this lesson students will be able to:

- Understand the properties and dangers of carbon monoxide (CO), explain how it is produced, and recognize its impact on human health.
- Identify the types of environmental sensors used to detect gases and temperature.
- Graph sensor readings and determine when an environment becomes unsafe.
- Design a simple alert system using Arduino, LEDs, and buzzers to respond to sensor data.

Challenge questions of the lesson.

- If you were designing this for a public space, what other features or types of alerts would you include to make the system accessible and effective for everyone?
- How could you adapt the code to handle different environments, such as a high-temperature kitchen? Why is it important for a safety system to have a backup plan in case a sensor fails?

SEL Relation

- **Self-Awareness:** The process of debugging code and troubleshooting hardware requires students to recognize and manage their own emotions, particularly frustration.
- **Social Awareness:** Designing the alert system with an accessible LED, buzzer, and screen encourages empathy and an understanding of diverse user needs.
- **Responsible Decision-Making:** Choosing the alert thresholds and designing a reliable system for safety requires students to think about the consequences of their choices.
- **Relationship Skills:** Working in a group to build and test the circuit or review each other's code helps students develop communication and collaboration skills.

Engage



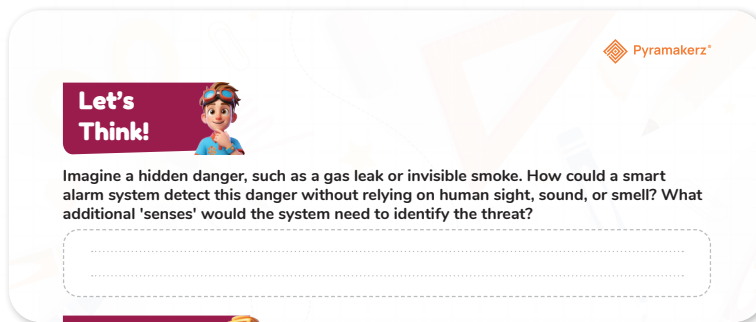
Teacher's Role:

- Spark curiosity about how devices detect hidden dangers.
- Connect to real-world applications of environmental sensors.
- Facilitate discussion and initial brainstorming.

Expected From Students:

- Participate in discussion, share initial ideas about how invisible dangers are detected.
- Engage with the "Think" question.

Parts of the book included.



Let's Think!



Objective:

Students will imagine and articulate how a smart alarm system could detect hidden dangers such as gas leaks or invisible smoke without relying on traditional human senses like sight, hearing, or smell. They will explore the concept of "extra senses" or alternative detection methods needed for such alerts.

Possible Answers:

- It would need special electronic "senses" called sensors.
- It would need sensors that can detect things like gas particles in the air, or changes in temperature.
- These sensors would act like artificial "noses" or "thermometers" for the system.

Possible Strategies for implementation:

- **"Invisible Dangers" Hook & Discussion:** Start with a hook: "In Lesson 1, we talked about silent dangers like gas leaks or carbon monoxide – things you can't see, smell, or hear. How do homes or factories detect these invisible threats? What kind of 'super senses' do buildings have?" Use this to transition into the "Think Question."

- **Teacher-Led Brainstorming:** Guide students to brainstorm ideas for how machines detect invisible changes, focusing on the concept of sensors. Use the brainstorm web to visually organize their ideas on the board.
- **Physical Component Introduction:** Briefly show the actual MQ-2 and DHT11 sensors. Point out their specific features. "Today, we're going to work with these specific 'senses' for our system: this is our 'robot nose' (MQ-2) and this is our 'robot thermometer' (DHT11). How do you think they send information?"

Explorer's Toolkit



Teacher's Role:

- Guide students in setting up their sensor circuits and uploading the code to the Arduino.
- Ensure students connect the gas and temperature sensors to the appropriate pins on the Arduino.
- Assist with troubleshooting during the setup and help students open the Serial Monitor to view sensor data.
- Encourage students to observe the readings and discuss how data can change based on environmental conditions.
- Emphasize the importance of monitoring real-time data for decision-making in sensor-based systems.

Expected From Students:

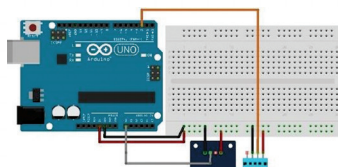
- Connect the gas and temperature sensors to the Arduino using the breadboard and jumper wires.
- Upload the provided code to the Arduino, ensuring it runs without errors.
- Open the Serial Monitor and observe the real-time data being displayed for both sensors.
- Experiment with environmental changes (e.g., heat, light) to see how sensor readings change.
- Record the data in their notebooks and note any significant changes.

Parts of the book included.

Explorer's Toolkit

DIY Lab

Connect the gas and temperature sensors to your Arduino, providing your future alert system with its "senses" to monitor the environment!
Upload your code and open the Serial Monitor to view the real-time sensor data.



DIY Lab

Objective:

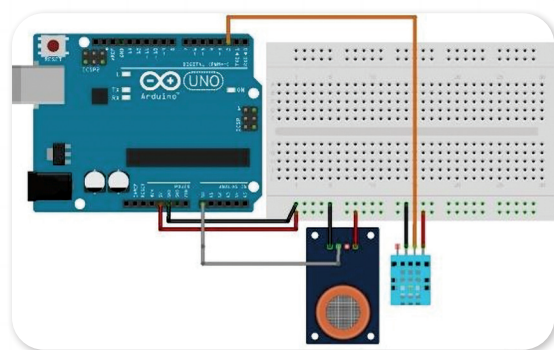
Students will connect gas and temperature sensors to an Arduino microcontroller to simulate an alert system that can detect environmental conditions. They will upload code and use the Serial Monitor to observe live sensor data, developing skills in hardware interfacing and real-time data monitoring.

Materials Needed:

- Arduino UNO board
- Breadboard
- MQ-2 Gas Sensor Module
- DHT11 Temperature & Humidity Sensor Module
- Jumper Wires (various lengths, different colors recommended for clarity)
- 220 Ohm Resistors (if DHT11 module doesn't include one; MQ-2 modules typically do)
- USB A-B Cable (to connect Arduino to computer)
- Computer with Arduino IDE installed
- Notebook or science journal for recording observations
- For Teacher-Led Demonstration: Small fan or cool/warm object (e.g., a cup of warm water, not hot) for DHT11, and an unlit butane lighter or similar very mild gas source for controlled demonstration of MQ-2 response, conducted in a well-ventilated area.

Activity Instructions:

- **Hardware Setup**
 - Connect the MQ-2 gas sensor to the Arduino:
VCC to 5V
GND to GND
A0 (analog output) to A0 pin on Arduino.
 - Connect the DHT11 temperature sensor:
VCC to 5V
GND to GND
Data pin to digital pin 2 (or any available pin).
- **Upload the Code**
 - Open the provided Arduino code in the IDE.
 - Upload the code to the Arduino board.
- **Open the Serial Monitor**
 - After uploading the code, click on the Serial Monitor in the Arduino IDE to see the data output in real time.



- **Observe and Experiment**
 - Notice the sensor readings. Ask students to test how the readings change by blowing air near the gas sensor or warming the temperature sensor by holding it in their hands.
- **Record Observations**
 - Students should record any changes in data in their notebooks.

Possible Strategies for Implementation:

- **Pair Work:** Have students work in pairs to set up the circuit and troubleshoot any issues together.
- **Guided Reflection:** Ask students to reflect on what happens to the sensor readings when they change environmental conditions (e.g., blowing air near the gas sensor).
- **Interactive Troubleshooting:** Walk around the class to help students with wiring or code errors, offering real-time feedback and corrections.
- **Extension Challenge:** Encourage students to design an experiment where they test how a specific environmental factor affects sensor readings, such as varying light or introducing different gases.

Explain



Teacher's Role:

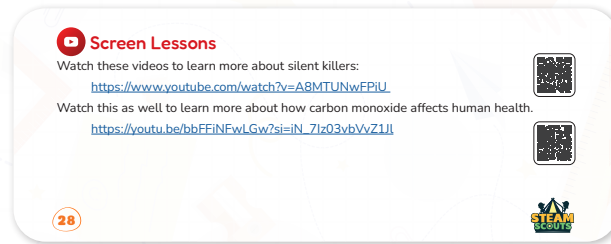
- **Content Guide:** The teacher introduces the article about how sensors work and guides students to the key concepts.
- **Discussion Facilitator:** The teacher leads a discussion to clarify the article's content, ensuring students understand that sensors convert stimuli (like heat) into electrical signals.
- **Contextualize:** The teacher helps students connect the technical information from the article to real-world applications, such as how sensors are used in robots to detect heat, motion, and carbon dioxide.
- **Explains Health Risks:** The teacher specifically explains the concept of "silent killers," focusing on how carbon monoxide (CO) is a colorless, odorless gas that can be deadly. They discuss the sources of CO and the importance of using alarms for safety.

Expected From Students:

- **Active Reading:** Students are expected to read the article to understand how sensors function, including how they rely on electrical circuits and conductors.
- **Information Synthesis:** Students should be able to explain how a sensor converts a physical condition (like temperature) into an electrical signal that can be processed.
- **Critical Thinking:** Students are expected to think about and discuss the practical applications of sensors in various scenarios.

- **Knowledge Application:** Students should be able to use the information they've learned to contribute to a design or problem-solving task.

Parts of the book included.



Screen Lessons

Objective:

Students will learn about silent killers such as carbon monoxide by watching educational videos and deepen their understanding of sensor technology through reading an article on how sensors work.

Resource Links:

Silent Killers – Learn about invisible, harmful gases found at home:

<https://www.youtube.com/watch?v=A8MTUNwFPiU>



How Carbon Monoxide Affects Human Health – Discover how this gas impacts the body:

https://youtu.be/bbFFiNFwLGw?si=iN_7Iz03vbVvZ1Jl



Possible Strategies to Implementation:

1. Pre-Watching:

- Ask students what they know about “invisible dangers” in the home and list their ideas on the board.

2. During Watching:

- Instruct students to note any causes, effects, or prevention methods mentioned in the videos.

3. Post-Watching:

- Facilitate a short discussion: “Why is carbon monoxide called a silent killer?”
- Have students create a simple safety poster or infographic showing ways to reduce the risk of gas exposure.
- Encourage reflection: “What changes can we make at home to stay safe from harmful gases?”



Dig Deeper

Objectives:

Students will be able to explain the fundamental principles behind sensor function and how they convert physical conditions into electrical signals. This understanding will enable them to analyze and apply this knowledge to a design problem, such as integrating a

sensor into a product to monitor its environment.

Article link: <https://www.sciencelearn.org.nz/resources/1602-electricity-and-sensors>



Possible strategies to implementation:

1. Pre-Reading: Discussion.

- **Strategy: Prior Knowledge Activation.**
 - Implementation: Ask: "We've seen our sensors work and watched videos about dangers. Now, let's dive into the science behind how these sensors actually convert physical changes (like gas presence or temperature) into electrical signals. What do you already know about electricity and sensors?"

2. During Reading: Engagement.

- **Strategy: Guided Reading & Annotation.**
 - Implementation: Have students read the article "Electricity and Sensors". Encourage them to highlight key definitions (e.g., transducer, resistance), underline important sentences about how sensors work, and make notes in the margins.

3. Post-Reading: Summarize and reflect.

- **Strategy: Summarize Key Learnings & Q&A.**
 - Implementation: Ask students to summarize the main idea of how a sensor works in their own words. Conduct a whole-class Q&A session: "What is a transducer in simple terms?" "How does a sensor use electricity to measure something?" "What was the most interesting technical detail you learned about how sensors like ours work?" Address any remaining questions or confusions from the reading.

Elaborate



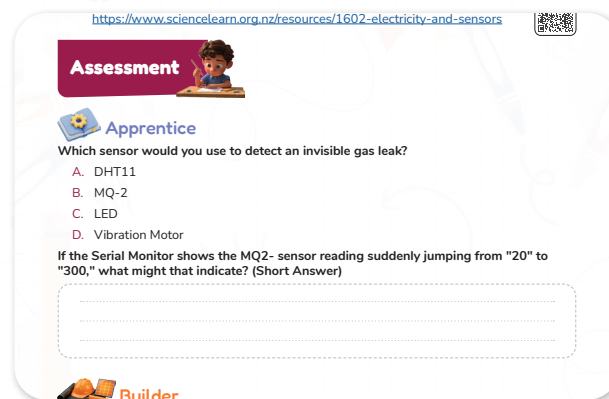
Teacher's Role:

- Support students in connecting prior knowledge about sensors, coding, and Arduino to the real-world challenge of detecting gas leaks.
- Guide students through the Apprentice and Builder activities, ensuring they understand both the technical content (sensors, threshold settings) and safety considerations.
- Encourage critical thinking by asking "what-if" questions to explore implications of their decisions.
- Differentiate the level of scaffolding: provide direct hints for Apprentice questions if needed, while allowing open-ended exploration for Builder.
- Facilitate small group or partner discussions to deepen understanding of how sensor data can be interpreted and acted upon in real-world contexts.

Expected From Students:

- Complete the Apprentice task by applying knowledge of sensor types and interpreting sensor readings.
- Use reasoning to explain why a specific sensor is appropriate for detecting a gas leak.
- In the Builder task, define meaningful “danger” thresholds for both gas and temperature readings.
- Apply creative thinking to propose system responses when danger is detected, considering both safety and practicality.
- Engage in peer discussion to compare and justify chosen thresholds.

Parts of the book included.



Apprentice

Objective:

Students will identify appropriate sensors for detecting invisible gas leaks and interpret sensor data readings from the MQ-2 sensor. They will develop understanding of sensor function and data analysis related to environmental monitoring.

Answers:

• Question 1

b. MQ-2. This is a gas sensor specifically designed for this purpose. The other options are a temperature/humidity sensor (DHT11), a light-emitting diode (LED), and a vibration motor (actuators, not sensors).

• Question 2: Short Answer

Expected Answer: A sudden jump from 20 to 300 on an MQ-2 sensor reading likely indicates the presence of a flammable gas leak. The significant increase in value corresponds to an increase in the concentration of the gas in the environment.

Possible strategies to implementation:

- **Think–Pair–Share:** Have students reason out their answers, then share and defend them.
- **Evidence Connection:** Ask students to link their choice of sensor to prior activities or lessons on sensor function.
- **Scenario Simulation:** Pose a “gas leak in the lab” scenario and discuss immediate steps.
- **Misconception Check:** Address common confusion between temperature/humidity sensors (DHT11) and gas sensors (MQ-2).



Builder

Objective:

Students will define threshold alert levels for gas and temperature sensors to make their sensor system “smart.” They will determine what sensor readings indicate danger and design appropriate system responses for detecting hazardous conditions.

Materials:

- Arduino with connected MQ-2 sensor and DHT11 (or similar)
- Computer with Arduino IDE
- Optional: LED, buzzer, or vibration motor for output simulation

Activity instructions:

- Recap what the MQ-2 and DHT11 sensors measure and typical reading ranges in safe environments.
- Introduce the concept of a “threshold” in sensor-based systems.
- Guide students to choose a threshold value for gas concentration and temperature.
 - Example: Gas > 250 ppm, Temp > 40°C.
- Ask students to decide what the system should do when a danger threshold is reached (e.g., activate buzzer, send alert, cut off power).
- Have students document their thresholds and system responses in their engineering journals.
- Invite volunteers to share their thresholds and justify their reasoning to the class.

Possible Strategies for Implementation:

- **Real-Data Analysis:** Show sample data logs and discuss which values could signal a hazard.
- **Collaborative Decision-Making:** In groups, students negotiate a final set of thresholds and actions.
- **Safety First Discussion:** Link choices to safety protocols in industrial and household contexts.
- **Prototype Planning:** Encourage students to think ahead to how they might code these thresholds into their Arduino program

Code:

```
#include <DHT.h>
#define DHTPIN 2      // Digital pin connected to DHT11 sensor
#define DHTTYPE DHT11 // DHT 11
#define MQ2PIN A0     // Analog pin connected to MQ-2 sensor
DHT dht(DHTPIN, DHTTYPE);
const int gasThreshold = 300;    // Gas level threshold
const float tempThreshold = 40.0; // Temperature threshold in Celsius
void setup() {
  Serial.begin(9600);
  dht.begin();
}
void loop() {
  // Read temperature from DHT11
  float temperature = dht.readTemperature();
  // Read gas sensor analog value from MQ-2
  int gasLevel = analogRead(MQ2PIN);
  // Check for sensor read errors
  if (isnan(temperature)) {
    Serial.println("Failed to read from DHT sensor!");
  } else {
    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.print(" °C\t");
  }
  Serial.print("Gas Level: ");
  Serial.println(gasLevel);
  // Check alert conditions and print warnings
  if (gasLevel > gasThreshold && temperature > tempThreshold) {
    Serial.println("ALERT: High gas level AND high temperature detected!");
  }
  else if (gasLevel > gasThreshold) {
    Serial.println("ALERT: High gas level detected!");
  }
  else if (temperature > tempThreshold) {
```

```
Serial.println("ALERT: High temperature detected!");  
}  
else {  
  Serial.println("All readings normal.");  
}  
Serial.println("-----");  
delay(2000); // Wait 2 seconds before next reading  
}
```

Evaluate



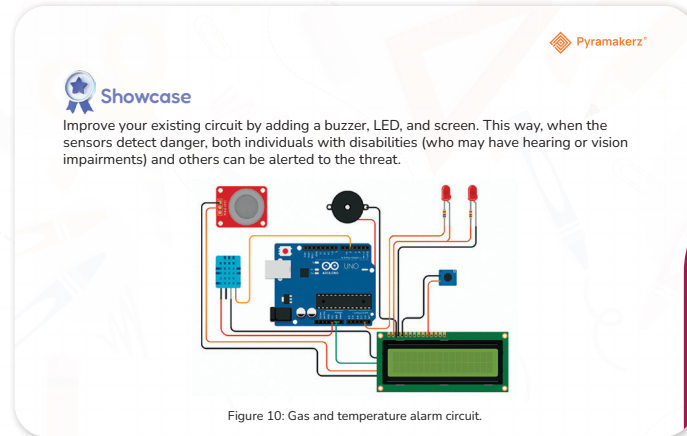
Teacher's Role:

- Guide students in enhancing their sensor circuit with additional output components for multi-modal alerts (buzzer, LED, screen).
- Emphasize inclusive design by discussing how different sensory outputs support individuals with varying abilities.
- Monitor wiring and coding for safety and accuracy, providing troubleshooting guidance when necessary.
- Facilitate peer demonstrations and feedback sessions, encouraging constructive critique and reflection.
- Assess student work using a clear rubric aligned with lesson objectives.

Expected from Students:

- Integrate a buzzer, LED, and LCD screen into their existing Arduino-based gas and temperature detection circuit.
- Program the circuit so that each component responds appropriately to danger thresholds (buzzer for audio alert, LED for visual signal, LCD for text warning).
- Demonstrate the working model to peers and explain how it addresses accessibility needs.
- Engage in peer review by observing others' projects and providing feedback on both functionality and inclusivity.

Parts of book included.



Showcase

Objective:

Students will improve their sensor circuit to create an inclusive alert system that visually and audibly communicates danger, ensuring accessibility for users with hearing or vision impairments.

Materials:

- Existing Arduino sensor circuit (with MQ-2 gas sensor, DHT11 temperature sensor, or equivalent)
- Buzzer
- LED (preferably red)
- LCD display module (16x2 or similar)
- Breadboard and jumper wires
- Arduino IDE with appropriate libraries for LCD and buzzer

Activity instructions:

1. Recap & Plan

- Review the thresholds set in the Builder activity.
- Discuss the importance of accessibility in technology design.
- Plan which outputs will be triggered by each threshold event.

2. Hardware Setup

- Connect the buzzer to a digital pin (e.g., pin 8) with ground.
- Connect the LED with a resistor to a digital pin (e.g., pin 9) and ground.
- Wire the LCD display using I2C or standard pins, ensuring correct SDA/SCL or RS/E/D4–D7 connections.

3. Code Integration

- Add logic to the Arduino sketch so that:

- Buzzer sounds when gas or temperature exceeds threshold.
- LED turns on as a visual alert.
- LCD displays a clear warning message (e.g., "GAS LEAK! EVACUATE").
- Include code comments to explain each section.

4. Testing & Debugging

- Simulate dangerous readings to check all components trigger correctly.
- Adjust thresholds or output responses as needed.

5. Demonstration

- Present the working system to peers.
- Explain how each component contributes to safety and accessibility.

Possible Strategies for Implementation:

- **Inclusive Design Discussion:** Facilitate a conversation on how multiple alert types ensure broader safety coverage.
- **Peer Collaboration:** Pair students with different strengths (coding, wiring, troubleshooting) for mutual support.
- **Scenario Simulation:** Create mock scenarios (gas leak, overheating) to test real-time responsiveness.
- **Iterative Improvement:** After initial testing, allow students to modify their design for faster or clearer alerts.

The code:

```
#include "DHT.h"
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
// --- Pin Definitions ---
// These constants define the Arduino digital pins connected to the sensors and outputs.
const int GAS_SENSOR_PIN = A0; // MQ-2 gas sensor is an analog sensor, connected to pin A0.
const int DHT_SENSOR_PIN = 2; // DHT11 sensor is a digital sensor, connected to pin 2.
const int BUZZER_PIN = 8; // Buzzer is a digital output, connected to pin 8.
const int LED_PIN = 9; // LED is a digital output, connected to pin 9.
// Define the type of DHT sensor being used.
#define DHTTYPE DHT11 // The DHT11 is used here. Other types could be DHT22 or DHT21.
const int GAS_THRESHOLD = 250;
// For the DHT11, a temperature above 40 degrees Celsius is considered high.
const int TEMP_THRESHOLD = 40;
```

```

// --- Object Instantiation ---
// Create a DHT object to manage the DHT11 sensor.
DHT dht(DHT_SENSOR_PIN, DHTTYPE);
// Create an LCD object for the I2C LCD display.
// The common I2C address for a 16x2 LCD is 0x27. Change this if your LCD has a different
address.
LiquidCrystal_I2C lcd(0x27, 16, 2);
// --- Setup Function ---
// The setup() function runs once when the program starts.
void setup() {
    // Begin serial communication at a baud rate of 9600.
    // This allows the Arduino to send data to the computer for students to monitor readings.
    Serial.begin(9600);
    // Initialize the LCD display.
    lcd.init();
    // Turn on the LCD backlight.
    lcd.backlight();
    // Print a welcome message to the LCD.
    lcd.setCursor(0, 0); // Set cursor to the first column (0), first row (0)
    lcd.print("Smart Alert System");
    lcd.setCursor(0, 1); // Set cursor to the first column (0), second row (1)
    lcd.print("Initializing...");
    // Initialize the DHT sensor.
    dht.begin();
    // Set the LED and buzzer pins as outputs.
    // This tells the Arduino that it will be sending signals to these pins.
    pinMode(LED_PIN, OUTPUT);
    pinMode(BUZZER_PIN, OUTPUT);
    // Add a delay to allow the sensors and LCD to warm up.
    delay(2000);
    // Clear the LCD after initialization message.
    lcd.clear();
    // Print a message to the serial monitor to indicate that the system is ready.
    Serial.println("Smart Sensor System Initialized!");
    Serial.println("Monitoring gas and temperature levels...");
}

```

```
}  
// --- Loop Function ---  
// The loop() function runs continuously after setup().  
void loop() {  
  // Add a small delay between readings to stabilize the sensor and avoid rapid updates.  
  delay(2000);  
  float temperature = dht.readTemperature();  
  int gasValue = analogRead(GAS_SENSOR_PIN);  
  // Check if either sensor reading is invalid (e.g., sensor not connected).  
  // The DHT library returns a NaN (Not a Number) if the reading is unsuccessful.  
  if (isnan(temperature)) {  
    Serial.println("Error reading from DHT sensor!");  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("Sensor Error!");  
  } else {  
    // Print the sensor readings to the serial monitor for student observation.  
    Serial.print("Temperature: ");  
    Serial.print(temperature);  
    Serial.print(" °C | ");  
    Serial.print("Gas Value: ");  
    Serial.println(gasValue);  
    if (temperature > TEMP_THRESHOLD || gasValue > GAS_THRESHOLD) {  
      digitalWrite(LED_PIN, HIGH); // Turn the LED on.  
      digitalWrite(BUZZER_PIN, HIGH); // Turn the buzzer on.  
      Serial.println("!!! DANGER THRESHOLD REACHED - ALERT ACTIVATED !!!");  
      // Update the LCD with a clear warning message.  
      lcd.clear();  
      lcd.setCursor(0, 0);  
      lcd.print("!!! DANGER !!!");  
      lcd.setCursor(0, 1);  
      if (temperature > TEMP_THRESHOLD) {  
        lcd.print("High Temp! Evac!");  
      } else {  
        lcd.print("High Gas! Evac!");  
      }  
    }  
  }  
}
```

```

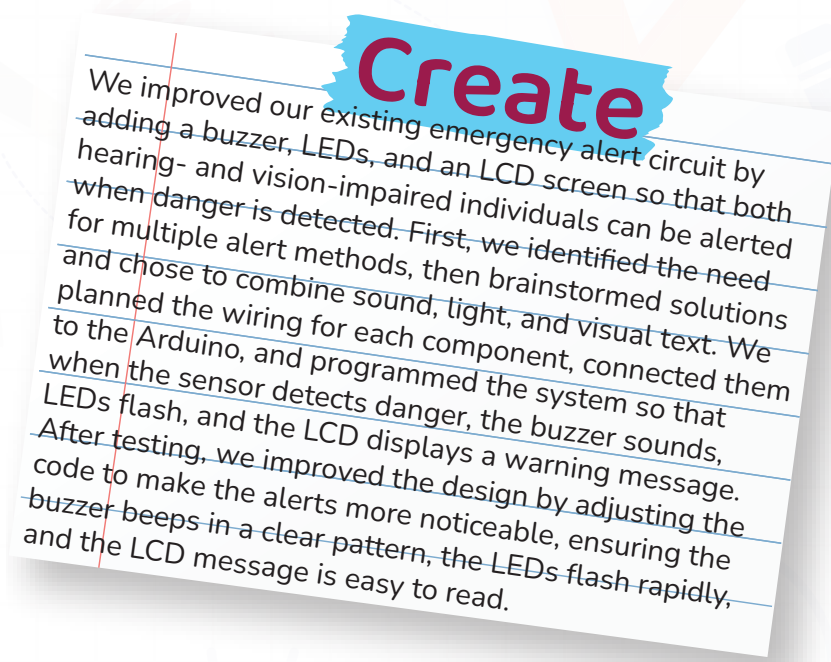
    }
  } else {
    // If conditions are safe, ensure the alert system is off.
    digitalWrite(LED_PIN, LOW); // Turn the LED off.
    digitalWrite(BUZZER_PIN, LOW); // Turn the buzzer off.
    // Update the LCD to display current sensor readings.
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Temp: ");
    lcd.print(temperature);
    lcd.print(" C");
    lcd.setCursor(0, 1);
    lcd.print("Gas: ");
    lcd.print(gasValue);
  }
}
}
}

```

Rubric – Accessible Alert System Showcase

Criteria	Needs Improvement (1–2)	Satisfactory (3)	Excellent (4–5)
Circuit Accuracy	Incorrect or unsafe wiring	Correct wiring with minor errors	Fully correct and safe wiring
Code Functionality	Code fails to trigger alerts	Code triggers alerts but with delays/inconsistencies	Code runs smoothly, all alerts function as intended
Accessibility Design	Only one type of alert present	Two types of alerts present	Multiple, well-integrated alerts supporting different needs
Creativity & Innovation	Minimal customization	Some unique features or enhancements	Highly creative design with thoughtful safety features
Presentation & Clarity	Incomplete or unclear explanation	Adequate explanation of system	Clear, confident, and thorough explanation with Q&A handling

It's time to move to the Create part and add this.



Now I Can

Objective:

Students will reflect on their learning journey throughout the lesson, identifying and articulating their mastery of key concepts related to carbon monoxide properties, environmental sensors, data analysis, and basic circuit design. This process facilitates metacognition and provides a foundation for self-assessment.

Activity Instructions:

Step 1: Revisit the Lesson's Work

The goal of this step is to jog students' memories and ground their reflection in concrete evidence of their work.

- Instruction to Students: "Take a few minutes to look back through your lab notebook, your sensor data graphs, and the code for your alert system. Review the 'Now I can...' statements and think about how your work connects to each one. Look for specific moments where you demonstrated that you understood a concept or mastered a skill."

Step 2: Student Self-Reflection

Students will now individually process their thoughts and document their learning. This can be done using a simple handout or an online form.

- Instruction to Students: "On your own, complete the following prompts. Be honest and specific in your answers. There are no 'right' or 'wrong' answers here; this is about understanding your own learning."

Step 3: Small Group Discussion

This step allows students to share their reflections and learn from their peers in a low-stakes environment.

- Instruction to Students: "In your small groups, take turns sharing your responses to the self-reflection prompts. As you listen to your peers, think about their experiences and how they might be similar to or different from your own. Remember to be respectful and listen actively."

Relation to Standards:

1. NGSS – Next Generation Science Standards

- **MS-PS4-3:** Integrate qualitative scientific and technical information to support the claim that digitized signals are a more reliable way to detect and communicate information than analog signals.
- **MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution.
- **MS-ETS1-2:** Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.
- **MS-ETS1-3:** Analyze data from tests to determine similarities and differences among several design solutions.

2. ISTE Standards for Students

- **1.4 Innovative Designer:** Students use a variety of technologies to create new, useful, or imaginative solutions to problems.
- **1.5 Computational Thinker:** Students develop and employ